NLP HW3 — Q2: WhoDunnit (Multi-Agent Systems)

Gal Barak

July 2026

2.1 Naive Agent Implementation (Analysis)

We ran the naive baseline with the default configuration and no additional flags:

```
python run_game.py --iterations 3
# 3 suspects, max_turn_count = 11, no summarizer, no internal thought
```

Here *Alex* is the Accuser and *Beth* is the Intel agent. The full log is reproduced in Appendix **??**; the analysis below cites it directly.

Behavioral problems identified (concrete examples)

Problem 1 — Analysis paralysis: the Accuser will not accuse even when the culprit is already uniquely determined.

- **Game 1** (culprit = Suspect 3). Only Suspect 3 wears square eye glasses. On the second question Beth answered `square eye glasses` → `Suspect 3`. Because the Accuser knows the culprit has square glasses, the culprit was uniquely identified at that point. Yet Alex asked *four more* redundant broad questions (long hair, black hat, red shoes, silver watch), never accused, and hit the turn limit.
- **Game 2** (culprit = Suspect 3). The answers `blue eye color` → `S2, S3` and `circular eye glasses` → `S1, S3` intersect to the single suspect {S3}. Alex again kept questioning and ran out of turns without accusing.

Problem 2 — Action-space collapse: the agent uses only one of its three actions. Across all three games Alex issued *only* `request-broad`: not a single `request-specific` question ever appeared. Consequently Beth was exercised only through `respond-broad`; her `respond` (Yes/No) action was never used. Both agents collapse onto a narrow slice of their available action space.

Problem 3 — No cross-turn reasoning / no candidate tracking. The Accuser treats every answer in isolation and never intersects the progressively narrowing suspect lists into a running "still-possible" set. This is the root cause of Problem 1: the agent literally cannot recognize that it is already done.

Do the agents use their full action space, or converge on one action?

They **converge**. The Accuser collapses onto `request-broad` (zero `request-specific` calls in all three games; `accuse` used only once, in Game 3), which in turn forces the Intel into `respond-broad` only. Because the terminal `accuse` action is under-used, information keeps accumulating but is rarely acted upon — so progress stalls and games tend to end in **timeout rather than a decision**. Games 1 and 2 both exhausted all 11 turns without an accusation.

Blind premature accusation, or endless loop until `max-turn-count`?

The Accuser gets stuck in an **endless questioning loop** — the opposite of a blind, premature accusation. Games 1 and 2 reached `max-turn-count` with no accusation at all; only Game 3 accused (correctly, Suspect 1), and even there only after redundant re-confirmation (`square eye glasses` → `S1` already sufficed, but Alex first asked about short hair and black hat).

Why it struggles to finalize:

1. **Stateless prompting.** Each turn re-reads the raw channel but is never guided to maintain or intersect a candidate set, so the agent never notices that a single suspect remains.
2. **No reasoning room.** The naive prompt explicitly forbids reasoning (*"not allowed to include any reasoning process"*), so the model defaults to the low-risk habit of asking one more question instead of committing.
3. **No firm stopping criterion.** "Accuse when certain" never triggers, because certainty is never established without candidate tracking.
4. **Sampling noise.** `temperature = 0.8` adds variability, but the dominant, consistent behavior is continued questioning.

A Naive baseline game log

```
Game 1
Suspects:
Suspect 1: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a black
     hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
      have long, black pants and a pink, button-up shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
     hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they
     have short, black pants and a pink, button-up shirt.
Suspect 3: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat,
      their hobby is dancing, is in a sad mood, they have red shoes, they have a silver watch, they have long,
      black pants and a green, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: green eye color
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: square eye glasses
Beth responded with: Suspect 3
Alex asked a broad question about: long hair
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a broad question about: black hat
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: red shoes
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: silver watch

=================
Game 2
Suspects:
Suspect 1: their eye color is green, they have circular eye glasses, their hair is short, is wearing a black
     hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
      have short, black pants and a pink, tee shirt.
Suspect 2: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat,
      their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have
     short, black pants and a green, tee shirt.
Suspect 3: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown
     hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they
     have short, brown pants and a green, tee shirt.
==
```

```
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: blue eye color
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: circular eye glasses
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: short hair
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a broad question about: brown hat
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: white shoes
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: silver watch

=================
Game 3
Suspects:
Suspect 1: their eye color is brown, they have square eye glasses, their hair is short, is wearing a black hat
    , their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they have
    short, black pants and a green, button-up shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
    hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have
    short, black pants and a pink, button-up shirt.
Suspect 3: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown
    hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a bronze watch, they
    have short, black pants and a pink, tee shirt.
==
The culprit is: 1
==
Communication channel:
Alex asked a broad question about: brown eye color
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: square eye glasses
Beth responded with: Suspect 1
Alex asked a broad question about: short hair
Beth responded with: Suspect 1
Alex asked a broad question about: black hat
Beth responded with: Suspect 1
Alex accused Suspect 1 and was correct!

=================
```